

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech (CL) Examination

March 2018

CL412 Professional Practices

Date: 28.03.2018, Wednesday

Time: 10.00am To 01.00p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1** Calculate quantities for concrete and steel work for R.C.C Beam shown in figure 2. **[07]****Q – 2 Attempt Any Two** **[14]**

- a. Refer figure 1 and calculate quantities for following items.
(i) Excavation (ii) P.C.C (iii) Foundation Masonry up to G.L (iv) Masonry above G.L
- b. Assume suitable rates and prepared abstract for items mentioned in Q-2 a.
- c. Calculate the quantity of plaster work for figure 1 and figure 2.

Q - 3 Attempt Any Two **[14]**

- a. Carry out rate analysis for first class brickwork in cement mortar (1:4).
- b. Draft the specification for 15 mm thick cement plaster in cement mortar (1:4).
- c. Find out rate of R.C.C Work for 1:2:4 per cum for slab.

SECTION – II**Q - 4** What are the types of estimate? Write difference between revised estimate and supplementary estimate. **[07]****Q-5 Attempt Any Two** **[14]**

- a. Determine the quantity of earthwork for a portion of a road from the following data:
The formation level of Road is 10.0m and the width is 7.50m. The side slope are 1:1(H:V) both in cutting and filling.

Ch.	0	30	60	90	120	150	180	210	240	270	300
G.L.	10.55	10.50	10.40	10.35	10.20	10	9.80	9.65	9.60	9.40	9.20

- b. Calculate the quantity of following items for slab culvert shown in figure 4.
a) Excavation b) Foundation Concrete c) Brick Masonry d) R.C.C work

- c. Calculate quantities of steel work for roof truss shown in figure 3. Consider the following details.

All members are single angles 50 x 50 x 6mm unless otherwise shown.

Weight of all bolts and rivets are 5% of total weight of truss.

All gussets are 6 mm thick. Base Plates are 12mm thick. Weight of a Rag bolt is 4kg.

Size of gusset plate at respective joints are as below

B 0.40 x 0.30

C, F,H 0.2 x 0.15

D,E 0.4 x 0.2

G 0.45 x 0.35

L,J 0.35x 0.3

K 0.45x 0.4

Q-6 Attempt Any Two

[14]

- a. Briefly describe the use of measurement book at site.
- b. Differentiate between Market Value and Book Value
- c. What are the different forms of Values and explain any three with example

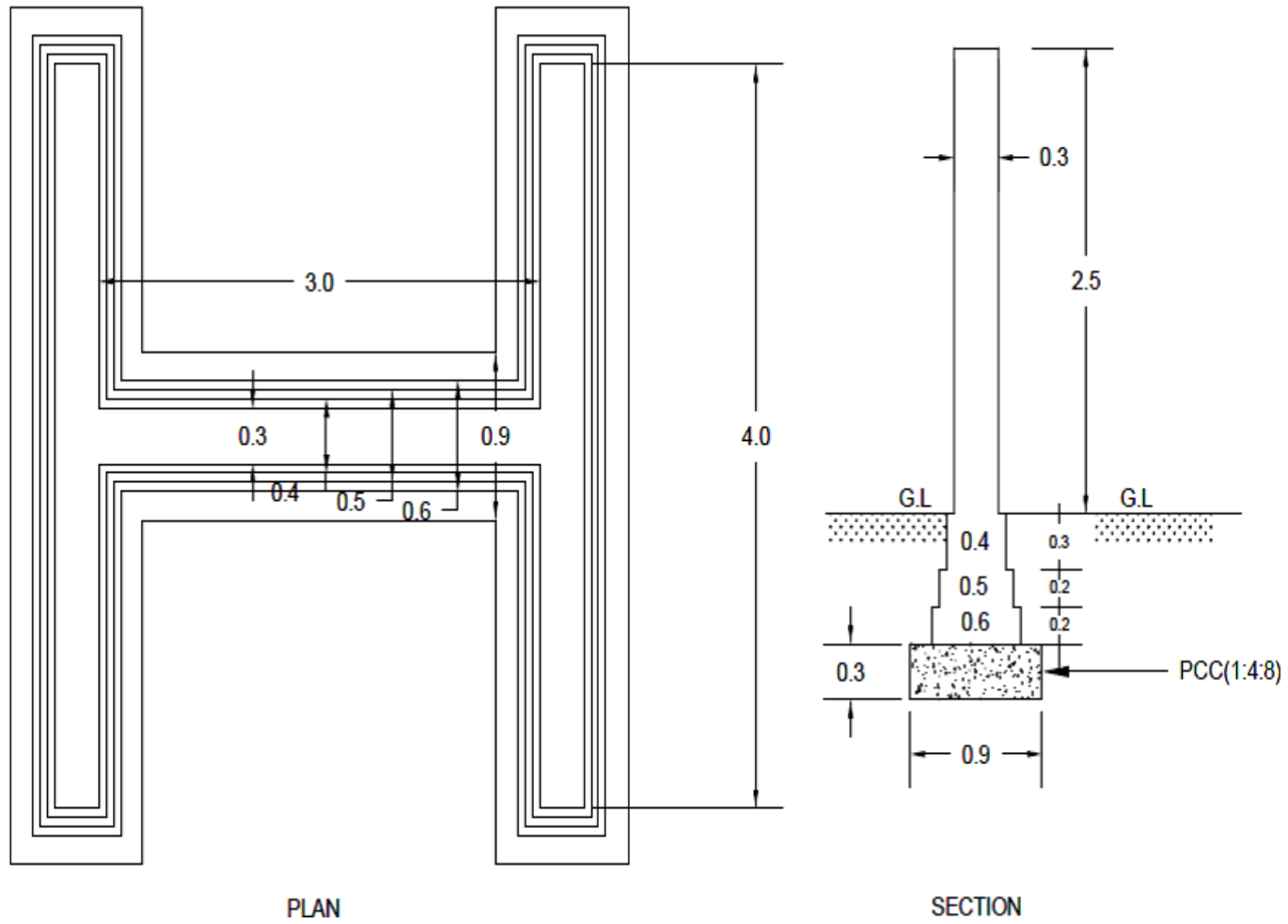


Figure 1
All dimensions are in meter

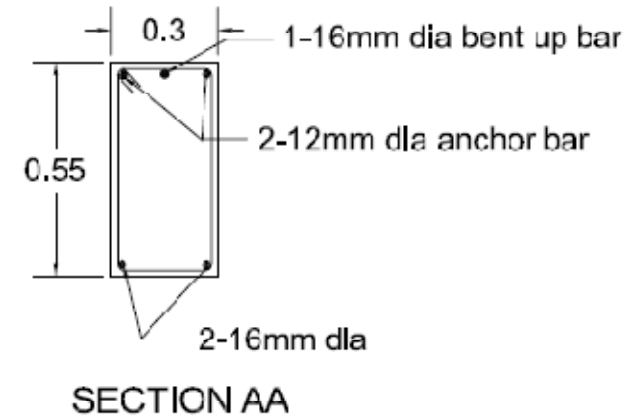
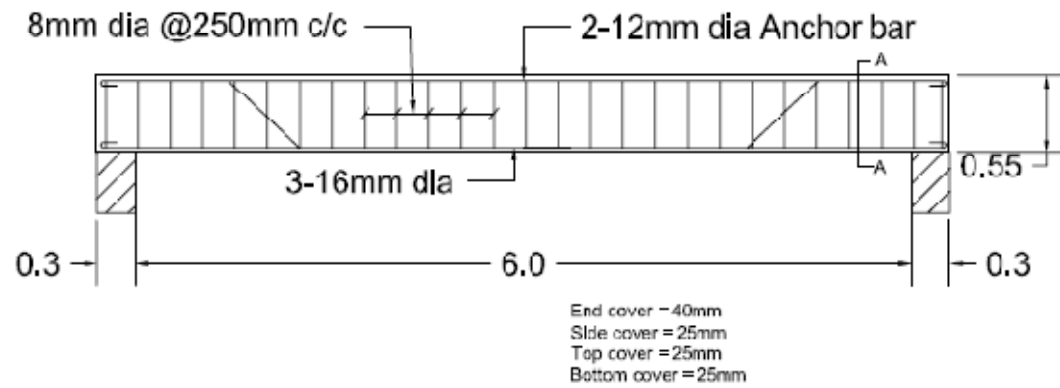


Figure 2.
All dimensions are in meter

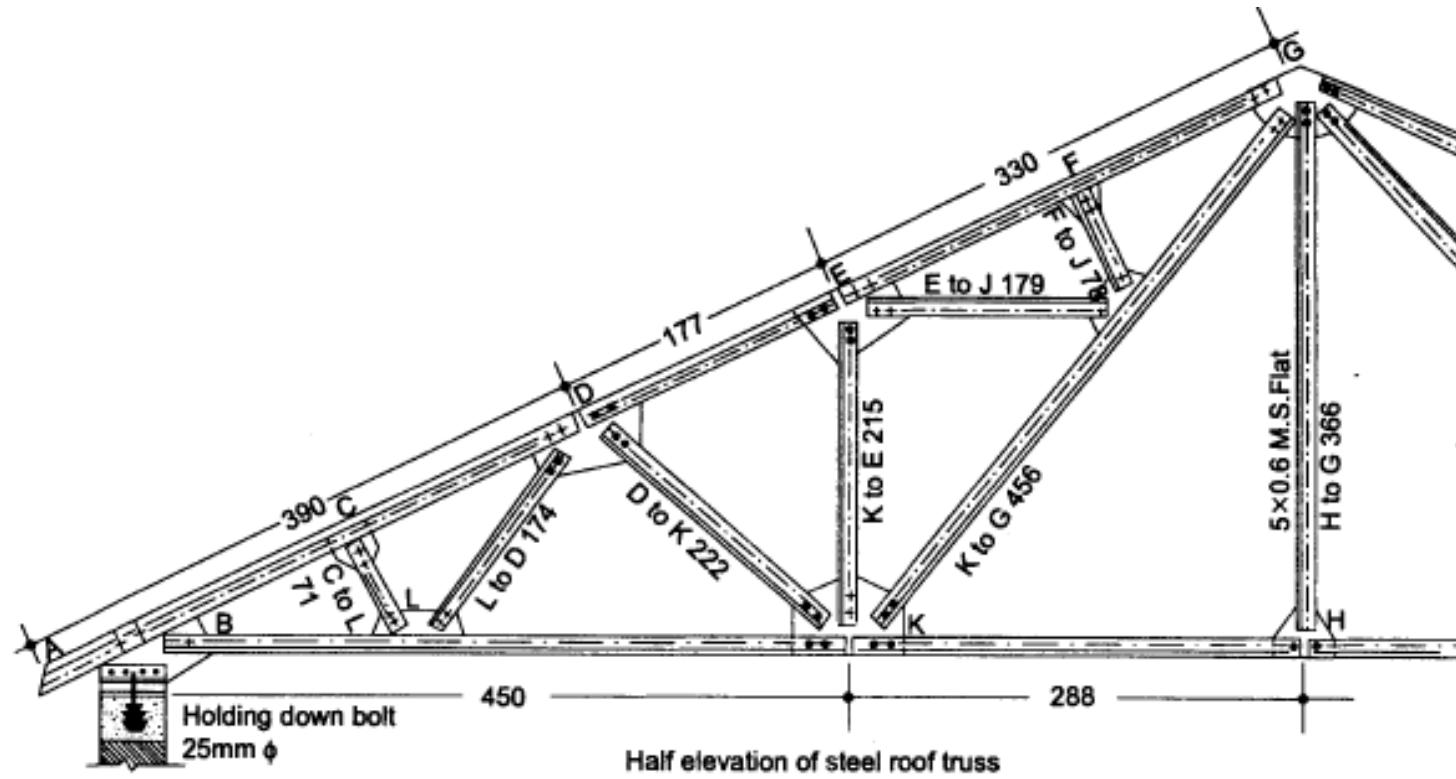


Figure 3.
All dimensions are in centimeter

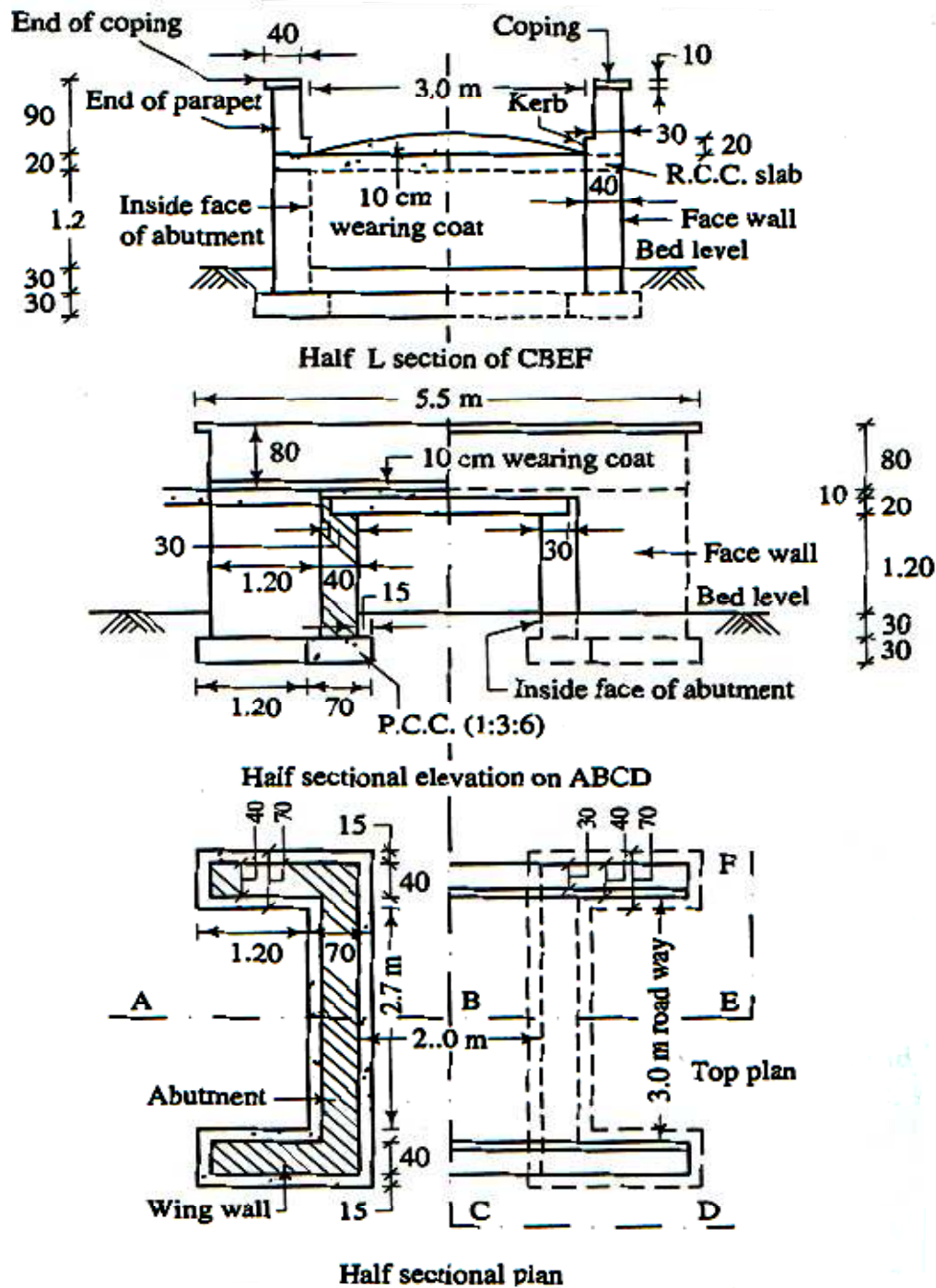


Figure 4.
All dimensions are in centimeter except
shown